Machine Learning Project Overview

DTSC691: Applied Data Science

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Project Overview:

The aim of this project was to create a model that accurately predicts the winner of the NCAA Men's Basketball Tournament, also known as March Madness. Data was gathered from multiple sources by downloading CSV files from each. Once the data was cleaned and formatted it was combined into a single data frame. New variables were created attempting to quantify the level of talent a team had going into the tournament, another metric that was created was an attempt to put a measurement on teamwork. The stated goal of the project was to get to a standard of 80% correct predicting of the score of a given match, this was not achieved for multiple reasons. Not the least of which was my inability to create a model that accounted for more than 40% of the variance in the scores. This was a very interesting project and there were a lot of lessons learned from it but will try and cover those as I work through the final overview.

Project Goals:

Purpose:

The purpose of this project was to be able to correctly predict the final team scores for each game of the March Madness Tournament, then using that to predict the winners of each game moving through the tournament. With the number of features that are accessible this day in age, it seemed like a straightforward task. Based on research that was conducted in the science of "bracketology", I thought this would be an interesting problem to try and tackle.

Project Focus:

One of the questions that I wanted to dig into was can you accurately predict winners without using actual statistics? Could I create metrics for things like teamwork or chemistry, a team's projected number of NBA draft picks, quality of roster, total combined player rank and have these metrics contribute to predictions in a meaningful way. I also attempted to create a metric for a team's probability of winning the tournament based on their season statistics.

Specific Goals:

The overall goal of this project was to achieve an overall correct prediction rating of over 80%. This metric was not achieved, with any combination of metrics, models or hyperparameters that were attempted.

Expected & Actual Outcomes

The expected result of this project heading into this project was that 1 or more of the player specific metrics would add value to the models. I ended up using Linear Regression, Linear Activation, Neural Networks, Random Forests and XGBoost models to create the models. I had expected to see that if a team was stacked with elite talent that they would make it further, or that if a team had balanced talent across starters and bench players that they may make it further in the tournament due to decreased fatigue.

Actual outcomes in terms of player metrics are that no one metric did better than bing relevant in more than 3 of the models, with none of them being relevant in the Random Forest Model. The basic linear regression had the fewest player features in its mix and the Linear Activation and Round Analysis sets both had 8. However, the Linear Activation Model had 3 of the 8 as dominate features being used in calculations. The only mode or set of variables that was dominated by player only metrics was the Round

Analysis where you can see lower values start to disappear as teams play into the later rounds.

Project Description

Project Objective and Scope:

In the sport of basketball there is truly no bigger event than March Madness, with over 80 million brackets being submitted every year. The bulk of which are made with many bias or incomplete data, the most common bracket type is that which the creator favors certain teams for a variety of reasons. Could be that it is their local team, alma marta, or their favorite team. Others go with their gut and how well they think a team is based on their record and a possible bias for how strong that teams conference is.

I had seen more elaborate models that took advantage of things like win streaks, proximity to home location or distance on the road, but I have never seen one with metrics regarding player rankings or projected draft location. I have also done some reading about how chemistry can make up for deficiencies in combined player levels, the greatest example of this was George Mason University in 2006 that started as the 11th seed, but others also 11th seeds include VCU in 2011, Loyola in 2018, UCLA in

2021 and NC State in 2024. This trend has become much more prevalent in the last 20 years as prior to 2006, it had only happened once in 1986 by LSU. The numbers are that in the first roughly 50 years it happened one time by LSU, then not again for 20 years and in the last 20 it has happened five times. I suspect that a teamwork metric would increase the percentage of correct score predictions, but the trick will be how this metric is created, meaning what statistics are used to produce the metric and what the formula is used.

Data Description:

Data will be gathered from multiple sources, additional offensive, defensive, advanced offensive, advanced defensive stats will come from Sports Reference College Basketball, individual college ratings and luck score will be gathered from Kenpom.com, team rankings to include tempo and injury ranks will come from Evanmiya.com. These sources all offer generally the same sets of statistics, but each offers a few that the others don't have, combined they will provide better footing for models to be constructed.

The sources for the individual player rankings and projected draft pick number will come from multiple sources ESPN Mock Draft and Research, CBS Sports Draft Prospect Rankings, NBADRAFT.net, Early Season Big Board and NBADraftRoom.com. Many of the draft projections are the same but they are different in other cases, getting multiple rankings for each team and then averaging them could make the metric better, but I am not expecting each teams value to shift to much, I expect that from site to site the first round will basically be the same with maybe 1 to 3 variations and even then I imagine that this will be them shuffled up or down 1 to 2 picks in either direction. The player rankings will come from CBS Sports Top 100 and Evanmiya.com rankings. Between the 2 sites Evanmiya I think will have the better data or at least the better statistics to put together.

One of the metrics that I would like to attempt to create a metric that measures teamwork or synergy of an individual team. I play to use information/statistics from the following sites to attempt to find a metric that works well. Presently I am not sure what information will get used for this portion of the analysis. Ultimately, I may end up creating multiple metrics and testing them against each other. To find a metric that preforms the best.

NCAA Scores and Tournament results will come from multiple sources, Yahoo Sports, Wikipedia, NCAA.com, sports-reference.com and ESPN. I will be looking for tournament year, round, winner, loser, winner_score, loser_score. Those are the pieces of data that I expect to need to utilize from these data sources. If I cannot

manage to scrape the data that I need from these as most of it is stored in tournament bracket format. Kaggle also published a March Madness Historical Data Set going from 2002-2025, this is a backup if I can't figure a way to scrap the data from one of the above sites.

Data Source Addresses

Sports Reference

1. https://www.sports-reference.com/cbb/seasons/men/2025-school-stats.html
   a. Years 2012 to 2025
2. https://www.sports-reference.com/cbb/seasons/men/2025-opponent-stats.html
   a. Years 2012 to 2025
3. https://www.sports-reference.com/cbb/seasons/men/2025-advanced-school-stats.html
   a. Years 2012 to 2025
4. https://www.sports-reference.com/cbb/seasons/men/2025-advanced-opponent-stats.html
   a. Years 2012 to 2025
5. https://www.sports-reference.com/cbb/seasons/men/2024-coaches.html
   a. Years 2012 to 2025
6. https://www.sports-reference.com/cbb/seasons/men/2024-conference-tournaments.html
   a. Years 2012 to 2025
7. https://www.sports-reference.com/cbb/postseason/men/2024-ncaa.html
   a. Years 2012 to 2025

Kenpom - Pomeroy College Basketball Ratings

1. https://kenpom.com/
   a. Years 2012 to 2025

Evan Miyakawa

1. https://evanmiya.com/?player_ratings
   a. Years 2012 to 2025

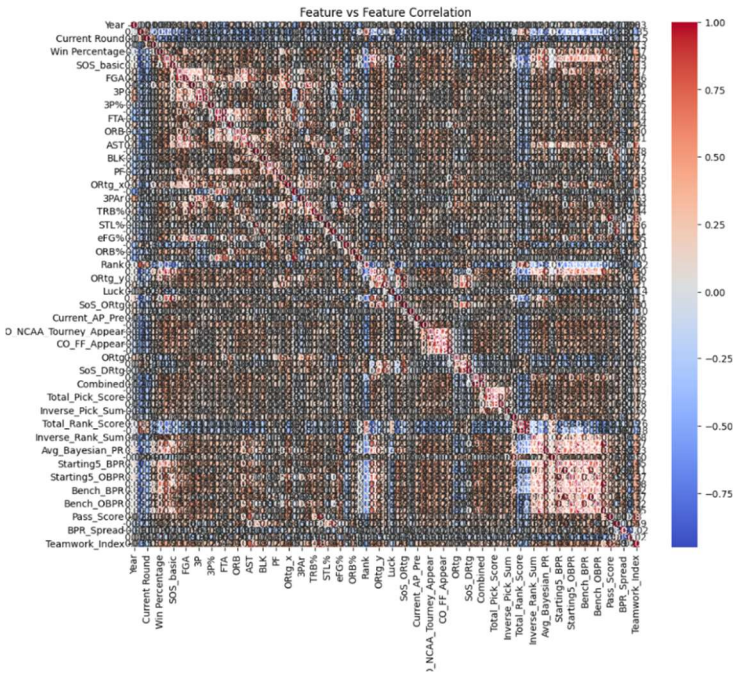
NBAMockDraft.net

1. https://www.nbadraft.net/nba-mock-drafts/
   a. Years 2012 to 2025

Kaggle

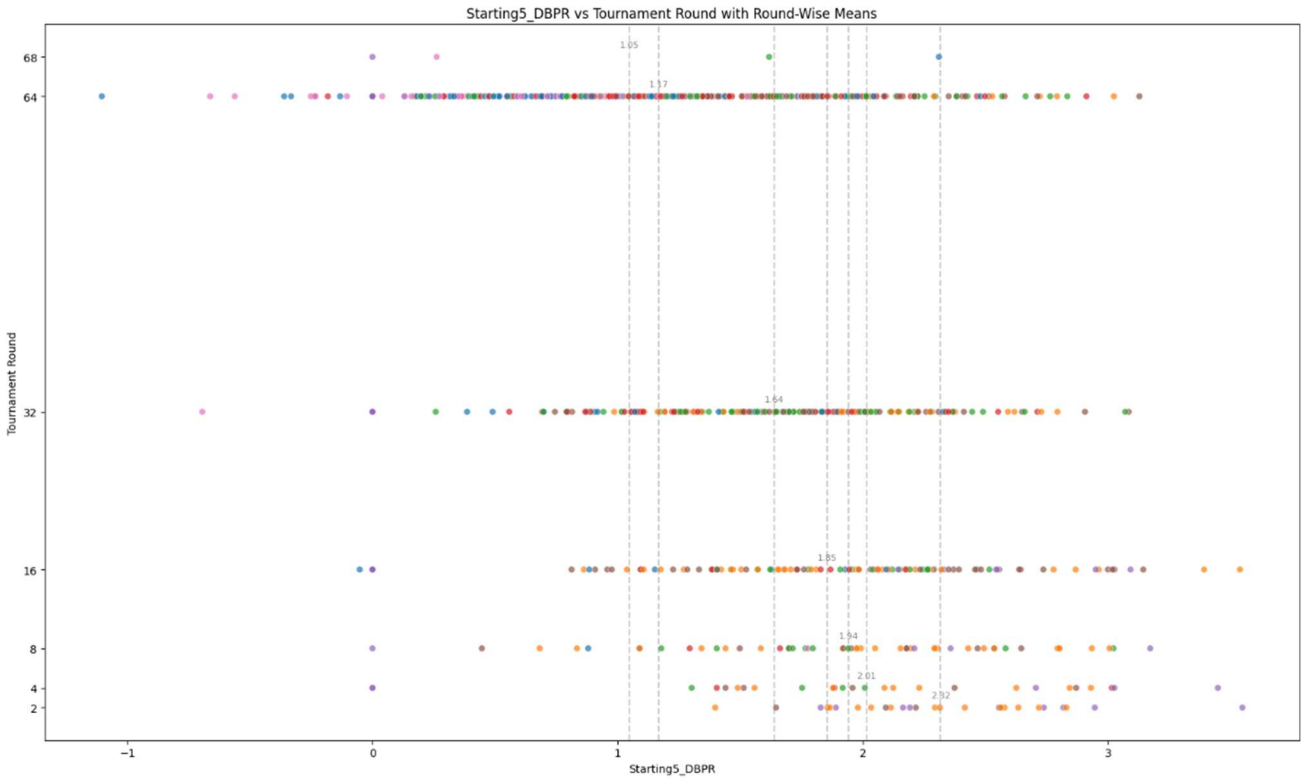
1. https://www.kaggle.com/datasets/nishaanamin/march-madness-datav
   a. Tournament Locations.csv
   b. Tournament Simulation.csv
   c. Tournament Matchups.csv
   d. 538 Ratings.csv

Exploratory Data Analysis:



Feature analysis was conducted starting with pairwise correlation, due to the number of features a basic heat map was used to see the different clusters. The values were then printed out so that the actual values could be observed. Next all features were run through a simple linear regression and were eliminated from the model in groups of 10 based on their F-score and p-values. I set the p-value at .05 for a .95 confidence interval and used this as the primary decision-making metric. I set the F-score to anything less than 0.9 due to the range of values that came out of the first ANOVA. There were 5 rounds of variable elimination before I saw a reduction in the R-squared value. I completed 1 last elimination leaving only the 19 features that fully met the thresholds I had set. This that elimination greatly reduced the R-squared value so it was determined that somewhere between 30-35 features would be needed to produce a good model using standard linear regression. Those features became the lin_reg_data_set.

After using the Anova to get an idea of how the data might stack up I wanted to perform a visual inspection of the data by round to see if there was a pattern, clustering or funneling of the values toward some point. It was found that as teams progressed the numbers tightened, showing that the talent level tightens or approaches a similar level as teams get closer to the championship game. This is supported by historical events that one and two seeds are the most predominant seeds in the final 4 and in the championship games, no team higher than a 8 seed ever making the championship game.



The features that showed funneling based on the mean for each round, were added to the rounds_analysis dataset. That would be tested with each model.

Variable analysis continued after each model was trained. Once the models were trained a permutation of variables was performed to determine feature importance for each of the models using the complete feature data set which was called bracket_data. The variables that showed influence both positive and negavtive were chosen to create the (model name)_data. These were then used in the next model that was tested.

(insert image of permutation of importance for a model)

Data Preparation and Cleaning:

This area was much more difficult that I had anticipated. Since I pulled data from multiple sources, my biggest hurdle ended up being naming conventions for all the different schools. As an example, there is Saint Francis (MD), St. Francis (MD), St Francis MD, St. Francis, St Francis, Saint Francis MD, there is also a Saint Francis in PA, but this give you an idea, a more basic was OSU, Oklahoma State, Ok State, Oklahoma St., etc. This made cleaning very difficult and locking down all possible ways to do something more difficult. I started trying to use just a fuzz reference and that got me to 85% complete but the issue ended up being that things like Alabama St would turn into Alabama since it matched over 85%. What ended up happening was

defining all possible division 1 college programs that have every played in the tournament, deciding on a standard name for each, then running fuzz at 95% after cleaning all the extra from the strings and creating an exception list for each college that did not get matched directly. The exception dictionaries ended up being very extensive with multiple schools having multiple aliases.

This issue caused me a lot of grief because it was in the bracket training code that I found the issue. Alabama State is played in the first four games but had been changed to Alabama, so it did not exist in the resource data statistics data frame. Teams that had St or St. had not been changed to say Ohio State so it was not returning their statistics for the simulation. Many of the errors happened due to my haste to move to the later analysis and modeling rounds but the St and St. issue happened because of how I read them as a person and not how the system would see it. My brain converted St to State, without thinking and it didn't occur to me that the code would not make the same connection. Ultimately, this was overcome but it did cost me multiple days, of coding and not making progress.

The second biggest issue was with one of the data sources information formating. Once it was acquired all the data was in a single column. The initial attempt to just transpose it was unsuccessful, with it not being able to do that simply. The data set ended up being broken down due to the empty cell between each entry. The other data sets were setup in a normal fashion so merging them was simpler. There were very few NaN values in the data and those that there were, typically were columns of NaN left for spacing purposes, and those columns were dropped.

Variable Training:

Many variables were created, as an attempt to quantify the quality of players on a given team. These features by their source data are as follows, from

- NBAMOCKDRAFT: Draft_Pick_Count, Total_Pick_Score, Avg_Pick_Number, Inverse_Pick_Sum
- Evan Miyakawa: Ranked_Player_Count, Total_Rank_Score, Avg_Rank_Score, Inverse_Rank_Sum, Avg, Bayesian_PR, Net_Eff_Dev, Starting5_BPR, Starting5_DBPR, Starting5_OBPR, Starting5_Plus_Minus, Bench_BPR, Bench_DBPR, Bench_OBPR, Bench_Plus_Minus

The EvanMiyakawa data tracked and ranked all 3000+ players in Division 1 basketball which is why there are so many player stats from that source. PR is Player Rating, DBPR is Defensive Bayesian Player Rating, OBPR is Offensive Bayesian Player Rating, Plus_Minus is a metric of how many points a players team scores when they are on the court verse when they are off the court. All Starting 5 metrics were created by

taking the top 5 players from each team and adding their specified metric to get a value. Bench metrics were created by taking players with rank 6 or greater and adding their metrics together. This was done to see if a team with a deeper bench would make a deeper run into the tournament or if teams with 5 superstars could carry a team deep into the tournament.
Other variables that were created were based on modeling if a team won the tournament or if they didn't, so one win person season. This was done with 4 modeling methods, linear activation classifier, neural networks classifier, random forests classifier and XGBoost classifier. A fifth metric was then created that average the output of the four models, this feature was called combined. These models produced a probability that given a teams end of season statistics that they would win the tournament. All the probabilities were very small, since of 64 teams there was only 1 winner each year. If I were to do this again, I would break this down further and go round by round, excluding seed from the equation as to not have it influenced the probability that a #16 could beat a #1 as I am sure that the seed feature would hold tremendous weight in those models. Those variables could then be retrieved separately as teams move through the rounds of the tournament. It also occurs to me that having calculated the player metrics first could have influenced the metric more or at a minimum from looking at the round analysis data help make better predictions on a round-by-round basis.

The final set of variables that were created are all related to the teamwork metric, but it is built on basic team statistics. Pass_Score, Def_Sync_Score, BPR_Spread, Scoring_Cohesion and Teamwork_Index. Each of the scores build on themselves and I had not expected any of them except maybe teamwork to influence the model. To create this metric, I weighed each of the metrics how I believed they might fit for a basketball team. So, each assigned weight is something that I decided on based on personal experience and knowledge. It occurs to me as I write this that this was not the best way to approach the metric. The different areas I believe are correct, scoring off of a pass shows teamwork, being able to stop a team from scoring shows defensive teamwork. But if I were to do this over again, I believe that I would use feature importance compared to points scored in each given round to produce the weights for each of the components. I think that would be a better way to have approached this variable in hindsight.

Model Training:

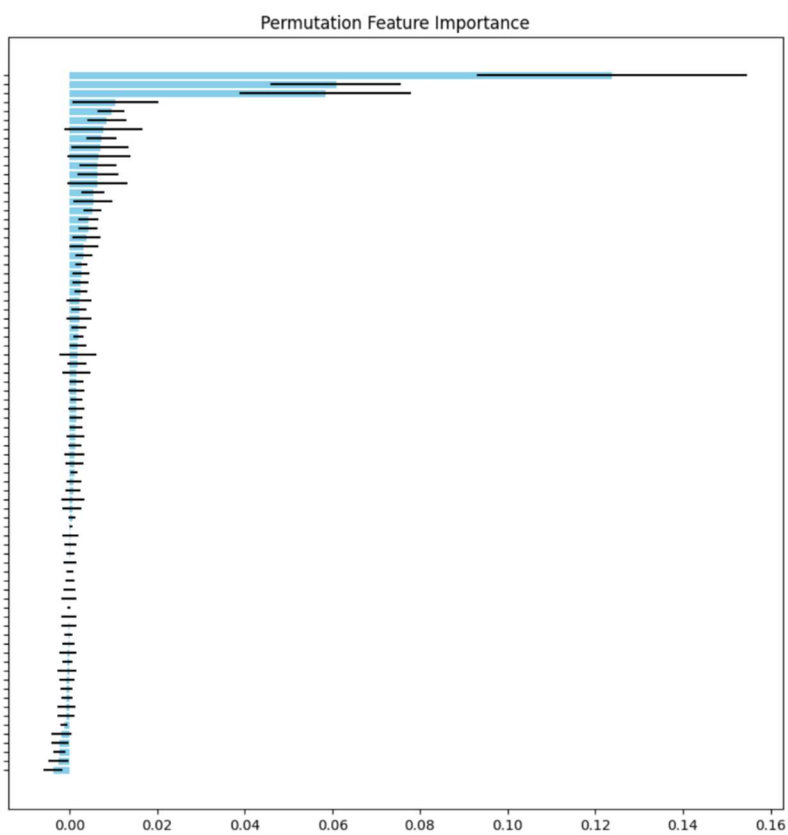
There were many lessons learned in this section of the project, many due to organization and how to keep track of what parameters had been tested and what should be kept and what should not. In the beginning the Linear Activation model was run and then copy and pasted with new parameters and this process was repeated, by the end parameters were notated as comments at the end of the model block with the best parameter notated, then that was left in the next iteration of parameters, until the

best set was found. Different scalers were added both Standard and Robust, along with different optimizers and schedulers.

Model	R-Squared Values												
Linear Activation	-2.956	-36.5857	-0.7408	-0.8993	-0.6356	-0.2911	-0.485	-0.9675	-0.2436	-0.3612			
Neural Networks	-1.629	0.3413	-0.6628	0.3118	0.3203	0.3606	0.3291	0.3306	0.361	0.336	0.3252	0.3377	0.2373
Random Forest	0.3968	0.4352	0.4715	0.3851	0.3441	0.285	0.223	0.0403	0.387				
XGBoost	0.3782	0.3057	0.2717	0.0936	0.0057	0.3687	0.3974	0.4046	0.4046				
	The solid black line represents the place that Opponent Score was taken out of the Bracket Data Set. It was carring a significant amount of variance, but is not something you would know before the tournament ever started.												

I feel this is where most of the lessons were learned, from different scalers, optimizers and schedulers that I had not ever tinkered with as well as using GridSearch to test multiple parameters in once iteration so that they could be compared more easily and efficiently. I did learn quickly not to try and test all my parameters all at once, as I tied up both my GPU and CPU for hours doing that at first.

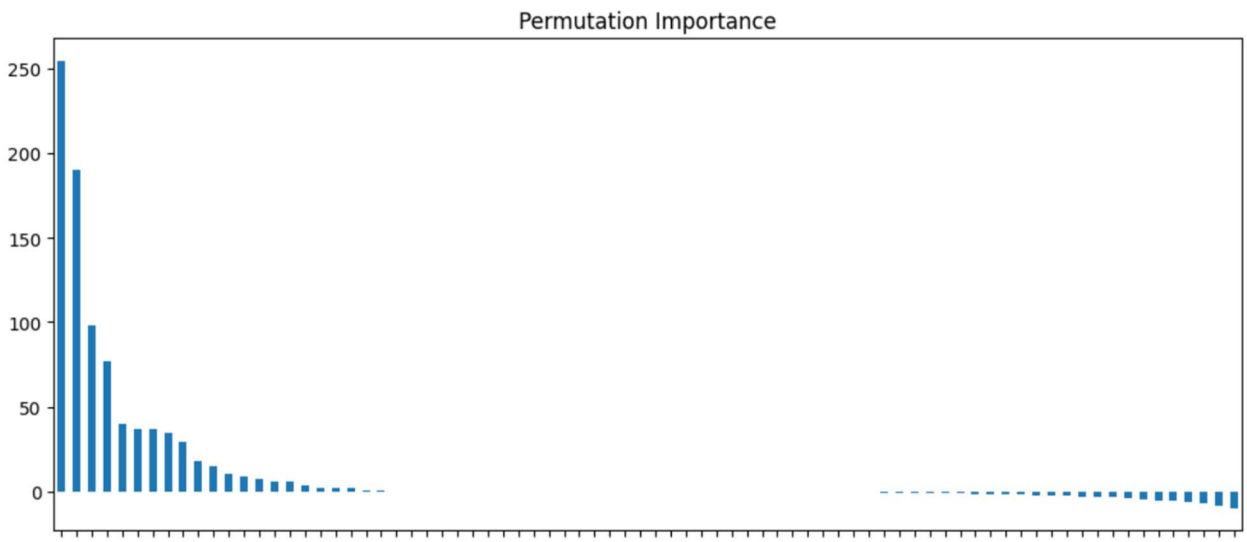
I trained models for Linear Activation and Neural Networks using PyTorch and I trained Random Forest and XGBoost models with Sklearn. They were trained in the following order Linear Activation, Neural Networks, Random Forest and XGBoost, with each model training utilizing lessons learned from the previous model build. Ultimately XGBoost produced the best results, though not near what I had hoped they would be with a best R-squared score of .405, the Random Forest Model was the next best modeling option with between .37 and .38 R-Squared.



After building the models and testing different hyperparameters, scalers, optimizers and schedulers for the different models. Variable permutation analysis was conducted for each of the model types to see which variables they responded to the most.

In outputs except the Random Forest model, at least three of the player metrics were included as having impact on the model itself to various degrees.

As stated earlier to of the models had 8 player metrics, but the it was clear in the Linear Activation model due to the clear tiers in it feature permutations that 3 of the player metrics greatly influenced the model.



Bracket Building:

The next set of models that were created were based on the models that were trained. A basic model was created that simply calculated the $^{1}/_{Team\ Seed}$ for both teams, then added those values, concluding by dividing each team seed by that total to

Features	Variable Sets	Linear Regression Model	Round Variable Analysis	Linear Activation Model	Neural Network Model	Random Forest Model	XGBoost Model
Bench_BPR		x	x	x			
Bench OBPR		x	x				x
Starting 5 BPR		x	x				x
Draft Pick Count			x				
Total Rank			x		x		
Bench +/-			x	x	x		
AVG +/-				x	x		
Starting 5 +/-				x	x		
Total Pick Score				x			
Starting 5 DBPR				x			
Bench DBPR				x			x
Average Rank			x	x	x		
Inverse Rank			x				
Starting 5 OBPR							x

create what was used as the weight in random selection to determine the winner. This was done to simulate an entire tournament, once the base model was created the best performing models from each of the model types. These were then used to predict points scored in each specific game. The outputs for these brackets were not

always very accurate after the first 3 rounds. I had assumed from the onset that in the model training process the probabilities of each seed moving forward would be something the models would learn through training, since seed for both teams was a feature of the models. This assumption proved inaccurate, as all models exhibited high frequencies of #16 beating #1 and #15 beating #2, at one point in 10 simulated brackets a #15 or #16 seed won the tournament. Which as stated earlier neither seed has made it past the Sweet 16 in history so for them to win the whole thing consistently, pointed to 2 things. One, the models did put weight on the seed of each team but had not picked up on win frequency given what round they were in. Two, higher seeds from smaller or weaker conferences may have better statistics than the lower seeds from larger or stronger conferences. This leads to higher seeds knocking off lower seeds more often than has happened historically. It was for this reason that by round seed progression rates were built into the bracket simulators, which was something I had wanted to stay away from but seemed like the only way to achieve a realistic bracket from any of the models.

Model Evaluation:

Each of the models was evaluated based on their R-Squared values. Mean Squared Error was also calculated for each of the models in case two or more models had the same R-squared, but R-squared was the primary decision maker on which models were selected. This was also the deciding factor in regard to hyperparameters, optimizers, scalers and schedulers. What ever combination generated the highest R-Squared was the model that was kept. Ultimately, the final XGBoost model was chosen due to it having an R-Squared value of .405 and the next closest R-Squared was .37. It was that XGBoost model that was also chosen for use in the bracket generating function.

User-Interface Integration:

The User Application was completed with Flask and Python. The base template for how the application came from the resource video by JimShapedCoding. This template was then expanded upon to include all the required pages for the project: Bio, Resume, Projects, Game Simulator and Bracket Simulator. Bio, Resume, Project are the only have python code for their routes, while the Game and Bracket Simulators have multiple files. The first Simulator to be built was the Game Simulator. To create this the XGBoost model was pickled and then uploaded into the PyCharm Project and the dataset that the model was trained on was also uploaded into a SQLite3 Database. Both were accomplished in separate Python files, once they were uploaded, they were both introduced into the Game Simulator code so that both the model and team

features could be used to create predicted scores. So that the user had some control over the outputs four input variables were introduced to variability to the predictions. The first variable is called Chaos Level: It controls randomness, second was Seed Consistency: It controls how consistent teams play game to game, Model vs. History: determines how much historic seed win rates factor into the model and finally Normalize Scores: This determines with scores will stay within historical ranges. These were added for multiple reasons but 1 was so that the user could do more than just select the two teams and get an output score. If the user wants the underdog to win more then can adjust the Chaos level to 5, the Seed Consistency to 1.2 and then Model v. History to 0 and they will see some #16 seeds upsetting #1 seeds with some frequency. Where the Normalize Scores by round comes in most is keeping the scores in a realistic range. Without it you will see some negative scores or some 140 scores, ultimately the winner of the game is usually the same in either case given the same settings, but the score will change from iteration to iteration.

When the bracket simulator they have control of the same features, with exception to any team selection. The user makes their selection for the previously described features and runs the model. A complete bracket is generated all at once, with each round broken apart by the round headings. This will allow the user to work each separate round in the bracket. As they are filling them out for whichever bracket challenge or competition, they have it entered in.

Then biggest challenges that I faced were things I had not expected at any point in the process. First was that getting the model and the test variables to align correctly. My initial desire was to have all 5 of the top models simulate an output and have all five of them plus an average of them come out. This proved more difficult than anticipated, it also created a problem that I had not foreseen in that when working with the different modes and adding scalers to one model or tensors, it changed the test model so that test_bracket for the random forest model was different than the test_bracket for the XGBoost model. Once that was sorted, things moved more smoothly. The second was my code terminating after one simulation, which is obviously not what you want a game-by-game bracket simulator or a bracket simulator to do especially if you are going to create multiple brackets. This problem had a very simple fix that, I made more complicated than it needed to be. The final project I think looks clean and the bracket simulator produces historically accurate predictions.

Software:

With regards to python the two software products used most were Juypter Notebooks and PyCharm. Multiple packages were used to clean, manipulate, create, train, test,

visualize the data and models. Excel was used to visualize completed CSV files when they were needed and for the creation of quick tables for data reference. Microsoft Word was used for all of the writing and formatting of this paper; Google Docs was used for the proposal. Zoom was used to create the video walk through. I stated earlier that many packages were needed for python, the following list should cover most of them: pandas, numpy, PyTorch, Sklearn, os, random, requests, XGBoost, Statmodels, Matplotlib, rapidfuzz, cupy.

The software side was an area I had not anticipated learning so much, about how many more packages there were out there. I would also Google a question, and it would spit out a list of packages that would be helpful or could be used to complete the task. It truly showed me how much I didn't know or understand when it came to python.

Final Thoughts and Future Changes:

The goal of this project was to achieve an 80% prediction regarding the score of each team, this was not achieved or came close to. I attribute this to one big reason, lack of creative variables. As the project neared its end and gathering everything together for this overview, gaps or missed opportunities were seen. Where more variables could have been created that could have given more context or a better reading of how a game might go. An example of this could have been generating a set of margin variables where team1's stats are subtracted from team2's stats. This would have generated by how much better one of the teams were in each given area. It is the believe that this could have been a more useful set of metrics for prediction as it would change based on the teams playing in each game, each round instead of just using the same metrics for each team with really the only major change being the round number.

Continuing with variable creation, it was stated earlier with the probability of winning the tournament. It is believed that doing this with regard to winning each round would have been a better metric. It would also have generated better probabilities at each of the levels. It is also thought that hyperparameter testing for these models would have increased the accuracy of each of their respective models. Regarding scaling, both standard and robust scalers were tested and neither consistently outperformed non-scaling and in some cases was a step backwards. The thought of creating a set of variables that would be the z-score for each feature. As stated earlier, the teamwork variable setup could have been more technically refined.

Overall, this project was a lot of fun to attempt, to solve what I believe to be an unsolvable problem. There are multiple areas where it is felt that opportunities to improve the project were missed.

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